
Pixel Art Generation using a Tiny Diffusion Model

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Abstract

We train a class-conditional Denoising Diffusion Probabilistic Model (DDPM) on 89,400 pixel art sprites at 16×16 resolution across five categories, and extend it with Classifier-Free Guidance (CFG). We also evaluate a class-conditional GAN, which we do not pursue further due to its poor generation quality on this dataset. We conduct ablation studies on sampling steps, model size, and the effect of class conditioning, confirming that conditioning improves generation quality over an unconditional baseline. CFG is evaluated at two dropout rates and five guidance scales, but degrades FID across all classes — we think its due to insufficient model capacity at this resolution. These results imply that CFG may be more effective with larger models or higher resolutions. A nearest-neighbour analysis confirms no training memorization.

1. Introduction

Diffusion models excel at producing highly detailed and diverse images (Croitoru et al., 2023). However, most work focuses on continuous-tone photographic imagery, leaving pixel art largely underexplored. Pixel art features unique structural constraints such as hard edges, limited color palettes, low resolutions, and class-specific spatial layouts, meaning generic diffusion pipelines often yield blurred contours and poor stylistic consistency. (Li, 2025)

This project trains a class-conditional DDPM on the Pixel Art dataset (Elgazar, 2024), which contains 89,400 sprites across 5 categories, and extends it with Classifier-Free Guidance (CFG). We investigate three questions: (1) does class conditioning improve generation quality over unconditional diffusion, (2) does CFG further improve over standard conditioning, and if so, which classes benefit most,

and (3) does class imbalance affect generation quality across categories.

2. Related Work

(Ho et al., 2020) introduced DDPMs, framing image generation as iterative denoising via a learned reverse diffusion process. A forward Markov chain gradually corrupts data with small Gaussian noise steps, while a neural network is trained to predict and remove this noise at each timestep. By parameterizing the reverse transitions as conditional Gaussians and injecting timestep information via sinusoidal embeddings, the model effectively reconstructs clean images from pure noise.

(Ronneberger et al., 2015) proposed the U-Net architecture, which we adopt as our backbone due to its skip connections and suitability for spatial generation tasks.

(Ho & Salimans, 2022) introduced Classifier-Free Guidance (CFG), training a single model for both conditional and unconditional generation by randomly dropping class labels during training. At sampling time, the two predictions are blended via a guidance scale w , where higher values produce sharper, more class-faithful samples at the cost of diversity.

3. Method

3.1. Dataset

We use the Pixel Art dataset (Elgazar, 2024), a collection of 89,400 sprites sourced from an online game at a resolution of 16×16 pixels in RGB format. The dataset is divided into five categories with notable class imbalance, as shown in Table 1. This imbalance directly influences generation quality, as reflected in the per-class FID scores reported in Table 3 — underrepresented classes consistently yield higher FID, indicating greater difficulty for the model to learn their distribution. The data is normalized to $[-1, 1]$ and split 90/10 into training and validation sets using a fixed seed.

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3.2. Diffusion Process

We adopt the DDPM framework of (Ho et al., 2020). The forward process corrupts a clean image $\mathbf{x}_0$ over $T = 300$ timesteps using a fixed linear noise schedule, where β_t increases from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. Defining $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$, a noisy image at any timestep can be sampled in closed form:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (1)$$

The reverse process trains a network ε_θ to predict the added noise at each step, minimising:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\varepsilon}} [\|\boldsymbol{\varepsilon} - \varepsilon_\theta(\mathbf{x}_t, t, c)\|^2] \quad (2)$$

where c is the class conditioning signal. At inference, images are generated by iteratively applying the learned reverse step:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t, c) \right) + \sqrt{\beta_t} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3)$$

from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ down to $\mathbf{x}_0$, with no noise injected at the last step ($t = 0$).

3.3. Model Architecture

We use a Conditional Tiny U-Net (5M parameters) optimized for 16×16 resolution. The encoder-decoder backbone operates at three spatial levels: 16×16 , 8×8 , and 4×4 , using strided convolution for downsampling and transposed convolution for upsampling, with skip connections preserving spatial detail.

Each level contains a ResBlock with two convolutional layers, GroupNorm, and SiLU activations. Conditioning is injected via a shared vector $\mathbf{c}_{\text{cond}}$, combining a sinusoidal timestep embedding (Vaswani et al., 2017) to a learned class embedding (nn.Embedding). This vector is projected and broadcast spatially into each block’s residual path to modulate behavior. The bottleneck at 4×4 consists of two consecutive ResBlocks for additional global capacity. The final layer applies GroupNorm, SiLU, and a 1×1 convolution to produce $\hat{\boldsymbol{\varepsilon}}$ of shape $3 \times 16 \times 16$. The unconditional baseline uses the same architecture with the class embedding removed.

3.4. Classifier-Free Guidance

Following (Ho & Salimans, 2022), we implement Classifier-Free Guidance (CFG) to support both conditional and unconditional generation within a single model. During training, each class label c is randomly replaced

with a null token $\emptyset$ with probability $p_{\text{uncond}} \in \{0.1, 0.2\}$, using an additional embedding index (`NUM_CLASSES + 1`). The training objective remains standard MSE. At sampling time, two forward passes are performed per timestep and blended via a guidance scale w :

$$\hat{\boldsymbol{\varepsilon}} = \varepsilon_\theta(\mathbf{x}_t, t, \emptyset) + w \cdot (\varepsilon_\theta(\mathbf{x}_t, t, c) - \varepsilon_\theta(\mathbf{x}_t, t, \emptyset)) \quad (4)$$

Higher w produces more class-faithful samples at the cost of diversity, while $w = 1$ recovers standard conditional sampling. This shared architecture serves as both our conditional model and our unconditional baseline, ensuring a compact and fair comparison.

3.5. GANs – for comparison

Generative Adversarial Networks (GANs) (Goodfellow et al., 2014) consist of two networks: a Generator that produces images from noise, and a Discriminator that tries to tell real from fake. We included a class-conditional GAN as a brief baseline, but did not pursue it further. GANs are prone to mode collapse and training instability (Dhariwal & Nichol, 2021), and these issues are amplified on small, structured datasets like pixel art — where, unlike diffusion models, a GAN must capture the entire data distribution through a single fragile objective rather than a gradual, stable denoising process.

3.6. Training Setup

All models are trained for 100 epochs using AdamW (Loshchilov & Hutter, 2017) ($\text{lr} = 2 \times 10^{-4}$, weight decay = 10^{-4}), batch size 256, cosine annealing ($T_{\text{max}} = 100$), and gradient clipping (max norm 1.0). Experiments run on a Kaggle Tesla T4 (15.6 GB VRAM).

All models are trained under identical conditions and converge without overfitting, as shown in Table 2.

3.7. Ablation Studies

Sampling steps. As shown in Figure 3, generation quality degrades sharply below 100 reverse diffusion steps. At 25 and 50 steps, outputs unstructured noise. At 100 steps, coherent pixel art shapes emerge, suggesting this as the minimum viable step count. Steps 200 and 300 yield progressively cleaner results with negligible visual difference, confirming that 100 steps offers the best speed–quality trade-off.

Model size. We compare base channel counts $\{32, 48, 64, 96\}$ by comparing validation losses after 5 training epochs. Performance scales with size from 32 channels (1.33M parameters, loss: 0.1183) to 48 channels (2.86M parameters, loss: 0.0934). However, the 96-channel configuration (11.05M parameters, loss: 0.0606)

shows signs of underfitting, likely due to optimization difficulties within the restricted epoch budget. The selected 64-channel baseline (4.98M parameters) achieves a validation loss of 0.0116 at full convergence (100 epochs), confirming it provides the optimal capacity-to-parameter trade-off.

Conditioning. The conditional model outperforms the unconditional baseline trained under identical conditions, confirming that class conditioning provides meaningful guidance for sprite generation. The conditional model achieves a validation loss of 0.0116 compared to 0.0128 for the unconditional baseline, and produces visually more class-consistent outputs as also shown in Figure 4.

3.8. CFG Evaluation

We evaluate the impact of label dropout by training two CFG (Ho & Salimans, 2022) models with $p_{\text{uncond}} \in \{0.1, 0.2\}$ under identical conditions. At sampling time, predictions are blended via guidance scale w as shown in Equation 4, utilizing an extra embedding index ($\text{NUM_CLASSES} + 1$) for the null token. Table 3 reports per-class FID for the baseline and both CFG variants; Figure 5 and 6 show a visual comparison of both dropout rates.

To evaluate guidance strength, we measure class 1 FID across five scales $w \in \{1.0, 1.5, 2.0, 3.0, 5.0\}$ for both dropout rates. FID increases monotonically with w under both configurations: for $p_{\text{uncond}} = 0.1$, FID rises from 37.80 at $w=1.0$ to 55.64 at $w=5.0$; for $p_{\text{uncond}} = 0.2$, from 40.96 at $w=1.0$ to 55.50 at $w=5.0$. No scale outperforms the baseline (40.96), confirming that CFG provides no benefit for this model regardless of the choice of w .

Figure 7 shows FID versus guidance scale for both variants.

3.9. Metrics used

We evaluate generation quality with the Fréchet Inception Distance (FID) (Heusel et al., 2017), which compares real and generated images in a pretrained Inception network’s feature space. Lower FID means generated samples are closer to the real distribution.

3.10. GAN results

The GAN (1.22M params, 200 epochs, Adam) converged to a degenerate equilibrium — $\mathcal{L}_G \approx 0.80$ and $\mathcal{L}_D \approx 0.69$ throughout training, both approaching the theoretical random-guess value of $\log 2 \approx 0.693$ — indicating the discriminator failed to learn any meaningful signal, and explaining the poor visual quality of generated sprites compared to the conditional DDPM’s validation MSE of 0.0116.

4. Discussion and Conclusions

This project showed that a small conditional DDPM (4.99M parameters) can generate recognisable pixel art at 16×16 resolution, achieving a mean FID of 28.64 across five classes. Contrary to expectation, CFG degraded FID across all five classes under both dropout rates tested. With $p_{\text{uncond}} = 0.1$, mean FID rose from 28.64 (baseline) to 43.36, and with $p_{\text{uncond}} = 0.2$ to 40.93 — suggesting the degradation is not primarily driven by the dropout rate. Our conclusion to the limited model capacity at 16×16 resolution: the conditional and unconditional predictions might be too similar for guidance to reliably amplify class structure, causing the blending to amplify noise instead. These findings suggest CFG is better suited to larger models or higher resolutions, consistent with its success in large-scale systems such as Stable Diffusion (Rombach et al., 2022).

As a preliminary comparison, we briefly evaluated a class-conditional GAN on the same dataset. Despite GANs being competitive with diffusion models in certain settings, (Dhariwal & Nichol, 2021) demonstrated that diffusion models can surpass GANs in image synthesis quality — a finding reflected in our own results, where the GAN failed to capture the visual structure of the pixel art sprites, producing outputs that bear little resemblance to the training distribution, as shown in Figure 8. This motivated us to focus our main analysis on diffusion-based approaches — specifically the comparison between conditional, unconditional, and CFG variants — where generation quality and diversity are more reliably controlled.

Class imbalance in the dataset is only partially reflected in per-class FID scores. Classes 0 and 4 (8,000 images each) achieve the lowest FID (18.03 and 23.08 respectively), while class 1 (Fruits & objects, 32,400 images) yields the highest baseline FID of 40.96. This suggests that visual diversity and intra-class complexity matter as much as dataset size — fruits and round objects form a highly varied category that is harder to model despite having more training examples.

4.1. Nearest Neighbor Analysis

A nearest-neighbour cosine similarity analysis confirmed no memorization, with all per-class average similarities below 0.8, well under the 0.95 overfitting threshold. Results are shown in figure 9.

4.2. Limitations and Future Work

Limitations include the noisy nature of FID at 1,000 samples, the 16×16 resolution constraint, and the fixed linear noise schedule. Future work could explore cosine noise schedules or larger architectures where CFG guidance may be more effective.

4.3. Division of Work

Beyza Nur Elaskan was responsible for the base DDPM architecture, training pipeline, linear noise schedule, ablation studies on sampling steps, model size, and the effect of class conditioning, and the nearest-neighbour memorization analysis.

Marthe Elgawly extended the model with Classifier-Free Guidance (CFG), implementing modifications to both the training loop and sampling procedure, conducted the per-class FID evaluation pipeline and guidance scale analysis, and as a supplementary experiment implemented a class-conditional GAN on the same dataset to provide an architectural baseline beyond diffusion models.

Both members contributed to experimental design, result interpretation, and writing.

5. Statement on the Use of AI

AI tools (Gemini and Claude) were used as a consulting tool for the team. First it was consulted on checking whether the chosen topic matches the required guidelines or not - given the accepted topics, so we can understand the depth of the project, as the team wanted to work on a domain that neither of us had worked on before with no prior experience in this research topic *Image Generation*. It was also used to provide us with different ideas as we did not want to only implement something but rather build the code for the purpose of comparing different methods, in this case it suggested we build the CFG and the reference paper in which we followed for a fair comparison. In terms of coding, AI was used for debugging, fixing errors that we encountered along the way, plotting codes (not plotting the actual images but providing us with the code to plot the graphs needed), AI was used to explain theoretical concepts from the reference papers we have followed. It also was used to understand some outputs throughout the coding process so we can know if we have implemented something mistakenly or not. AI was used as a code refining tool for the GAN code as our main focus was on the diffusion but wanted to see how would the GAN results be. We wrote the pseudocode and AI provided us with the actual code to run. AI also provided us with guidance on the report structure and what to cover in each section with what level of detail. It was used at the end as a grammar check, text flow check, and to check whether we covered everything from our code or missed some details. AI was used to generate the equations in latex format for the report and code (markdowns added in the code) All submitted code has been fully run, verified, understood by both team members. All results, figures, conclusions, discussions are our own. AI was not used to generate the report but rather fix the english used.

Note: All referenced images and tables are included in the appendix.

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Appendix

Class	Category	Images
0	Characters (no weapon)	8,000
1	Fruits & round objects	32,400
2	Large fruits	6,000
3	Armors & arrows	35,000
4	Characters (with weapon)	8,000
Total		89,400

Table 1. Class distribution of the Pixel Art dataset.

Model	Train	Val
Conditional	0.0117	0.0116
Unconditional	0.0127	0.0128
CFG $p=0.1$	0.0120	0.0125
CFG $p=0.2$	0.0119	0.0116

Table 2. Final MSE losses after 100 epochs.

Class	Baseline	CFG $p=0.1$	CFG $p=0.2$
Chars (no weapon)	18.03	32.70	28.04
Fruits & objects	40.96	52.26	58.19
Large fruits	29.34	51.27	46.17
Armors & arrows	31.80	38.42	34.38
Chars (with weapon)	23.08	42.14	37.89
Mean	28.64	43.36	40.93

Table 3. Per-class FID (lower is better), CFG at $w = 5$.



Figure 1. Generated samples from the conditional DDPM, 10 samples per class after 100 epochs of training.

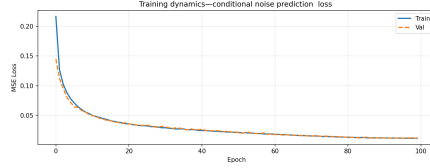


Figure 2. Training and validation MSE loss over 100 epochs.

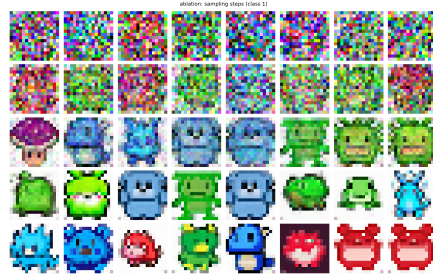


Figure 3. Ablation Steps - Class 1



Figure 4. Conditional (top) vs unconditional (bottom) generation, class 1 (Fruits & objects).



Figure 5. Baseline conditional (top) vs CFG $w = 5$, $p_{\text{uncond}} = 0.1$ (bottom), class 1 (Fruits & objects).



Figure 6. Baseline conditional (top) vs CFG $w = 5$, $p_{\text{uncond}} = 0.2$ (bottom), class 1 (Fruits & objects).

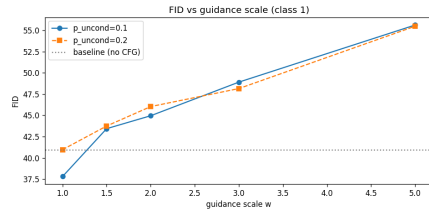


Figure 7. FID vs. guidance scale w for both CFG variants on class 1 (Fruits & objects). The dotted line marks the baseline conditional FID.



Figure 8. GAN conditional generation, 10 samples per category.



Figure 9. Nearest neighbour analysis (classes 0 and 1). All similarities below 0.53 confirm no memorization.